Who Bears the Load? Disparities in Childcare Time Across Family Structure, Gender, and Socioeconomic Factors in the U.S

Jasmine Chen '27 | Data Science Capstone Project

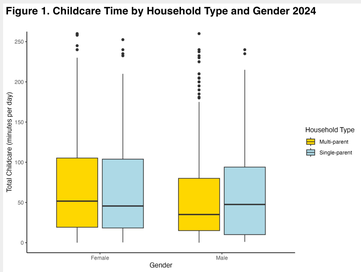
Background & Research Question

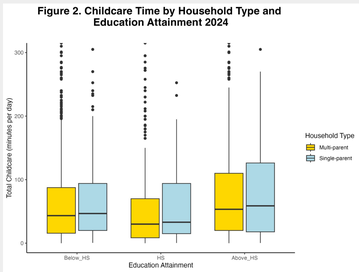
- Due to gender norms, stereotypes, and social expectations, the time parents spend on child care can vary based on family dynamics.
  - These variations can often imply underlying social values and biases in parental care.
- Child care is also a crucial factor to a child's growth and their physical and mental health. Based on previous literature (Kalenkoski et al. 2007), it is evident that family structure can have an impact on the time spent on child care.
- In this project, we attempt to use the **American Time Use Survey** to study the impact of family structure on time spent on child care in more recent days using newer data and compare whether similar trends are still observed, and whether there has been a change over time.

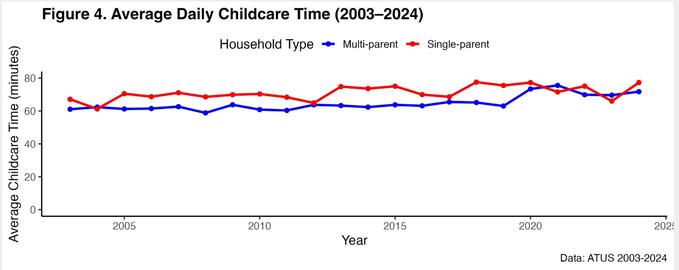
Research Question:

How does the **time spent on unpaid childcare** differ between **single parents** (households with only one adult and at least one child) and **parents with other members in their household** in 2024, and how does this relationship vary by **employment**, **gender**, **education**, and **region**? How has the trend changed over time since 2003?

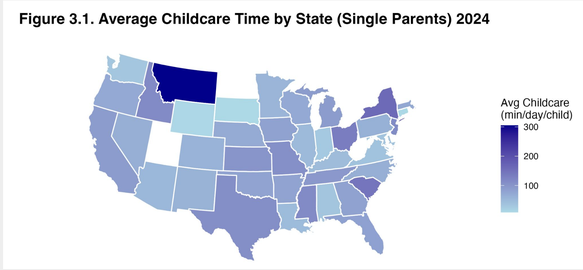
Visualizations

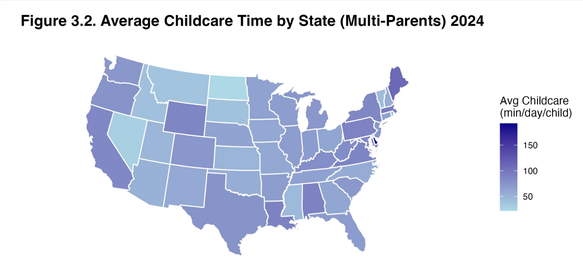






There is not a notable difference between parental structure with gender in average childcare time per child in 2024 (Figure 1). It seems like **parents with the highest education attainment** spend more time in childcare per child (Figure 2). There is not a drastic difference between single vs. multi parent households.





Montana has a drastically higher average for childcare time for single parents in 2024 (Figure 3.1). The same pattern is not observed for multi parents, where there is **a relatively even distribution** of average childcare time across the country (Figure 3.2). From 2003 to 2024, the average childcare time per kid for **single parent families** is **higher** than that for multi parent families for most years except for 2004, 2012, 2021 and 2023 (Figure 4). The trend between the two seems to be moving mostly simultaneously, with some fluctuations after 2020.

Methodology

Data

This study uses data from the American Time Use Survey (ATUS) conducted by the Bureau of Labor Statistics, combining the single-year 2024 file (n = 7,669) and multi-year 2003–2024 file (n = 252,808) to examine childcare patterns and trends over time.

Data Cleaning

- Single parents were identified as respondents with at least one own child and no spouse or partner in the household. The number of children was also calculated based on the roster file.
- Total daily childcare and work time were computed by summing activity durations from the activity file by activity code. Childcare time per child was calculated by dividing total childcare time by the number of children.
- Demographic variables including education attainment and region were merged in from the CPS file, and missing values were imputed with zeros.
- Final cleaned up data variables: **parental status** (4339 multi, 588 single), **gender** (2900 female, 2027 male), **number of children** (range from 1-7), **total childcare time**, **childcare time per child**, **work time**, **education attainment** (below, HS, or above), **state** (all 50 + DC)

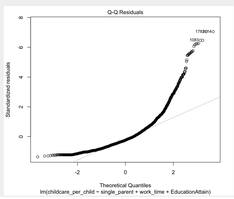
Modeling

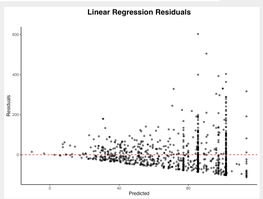
- A **VIF test** was conducted prior to model fitting to identify potential multicollinearity within predictors, and Gender and Region (6 specifically) were removed as a result in the MLR model.
- Two predictive models were estimated on the 2024 data using an **80/20 train-test split** with **k-fold cross-validation** (k = 5).
- Model 1: **Multiple linear regression (MLR)** with total childcare time as the outcome and single parent status, work time, and education attainment as predictors.
  - Constructed residual and normal Q-Q plot to verify model assumptions
- Model 2: **Random forest regression** (500 trees) using the same predictors plus gender and region.
  - Constructed residual plot for model performance evaluation
- Model performance was compared using **RMSEs** and **k-fold CV** scores on the held-out test set.

Models

Multiple Linear Regression Model

Variable	Estimate	Lower_95_CI	Upper_95_CI
Intercept	85.586	81.099	90.074
Single Parent (vs Multi-parent)	11.821	3.680	19.962
Work Time (minutes)	-0.094	-0.105	-0.083
High School (vs Below HS)	-8.410	-16.255	-0.565
Above High School (vs Below HS)	16.183	10.529	21.837

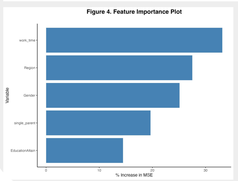


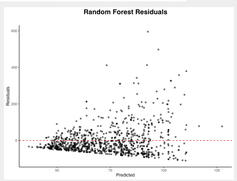


- **Major coefficient interpretations:**
  - On average, a single parent spends around 12 min longer per day on one child compared to a multi-parent, assuming all else equal.
  - There is a negative correlation between work time and childcare time, which is expected.
- **Evaluation Metrics:**
  - *RMSE*: 79.37631
  - *CV score*: 6807.125
- **Model Assumptions:**
  - Equal variance assumption likely not met from the pattern of the residual plots.
  - Normality assumption for MLR likely violated due to the large upper tail for the Q-Q plot.

Random Forest Model

	%IncMSE	IncNodePurity
Parental Status	19.638565571892	69326.8744359141
Work Time (minutes)	33.1538693326836	1482468.11954515
Gender	25.0990574713255	197498.154521164
Education Attainment	14.471540837276	236935.760726135
Region	27.4981011702026	1635499.30239667





- **Feature Importance:**
  - Work time is the most important feature/variable for prediction, followed by Region.
  - Education Attainment is the least important feature.
- **Evaluation Metrics:**
  - *RMSE*: 77.05554
  - *CV score*: 6930.476
- **Model Comparison:**
  - The CV scores suggest the MLR generalizes slightly better despite RF has a smaller test set RMSE (RF may be overfitting to the train split)

Conclusion & Discussion

Conclusions

- On average, **single parents seem to spend the most time** in childcare per child, which can be explained by the lack of support from other family members and the heavier parental duty as a single parent. This result also matches the previous findings.
- **Multi-family fathers** spend less time with their children compared to the moms. No notable difference between single fathers and mothers.
- **Parents with high school degree** seems to spend the least time in childcare for both groups, and there exists **some regional differences** in average childcare time, but these patterns are not the same for single- vs. multi-parent households. More Southern states seem to have greater average childcare time for single parents. Yet, there is not a definitive way to explain the regional variations.
- As expected, there is a negative correlation between childcare time and work time, suggesting that parents who spend more time in work spend less time with children.

Limitations

- The performances of both models are not ideal, with relatively high RMSEs, indicating high variance in the outcome. Two model assumptions are violated for the MLR model, and log transformation didn't resolve the pattern in the residual plot (will attempt again). Therefore, the model significance is affected, but the coefficients remain interpretable.
- Due to the lack of information, the way of defining single parents may not be fully accurate, and there are other factors that are not considered, including childcare support from other family members that don't live in the same household, intangible time on childcare, divorced/co-parenting parents, etc.

Future Work

- Future studies should focus on exploring the nuance between these other factors can our study cannot include to accurately classify single parents.
- Further studies need to be done to find the best models for predicting childcare time with more predictors.

Acknowledgement

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References

Kalenkoski, C. M., Ribar, D. C., & Stratton, L. S. (2007). The effect of family structure on parents' child care time in the United States and the United Kingdom. *Review of Economics of the Household, 5*(4), 353–384. https://doi.org/10.1007/s11150-007-9017-y